

Roll No.

TEC-591

B. TECH. (ECE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
TRANSDUCERS, ACTUATORS AND DISPLAY DEVICES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is the difference between Actuator and Sensor ? Describe various static characteristics of sensors. (CO1)

OR

- (b) Explain Thermoelectric effect. If the cold junction temperature is 3°C, neutral temperature is 275°C, find temperature of inversion. (CO1)

2. (a) Explain photoelectric effect. What is the difference between photoelectric and photoluminescence effect ? (CO1)

OR

- (b) The photoelectric work function of a material is 1.62 eV. Calculate the longest wavelength that can cause photoelectric emission from material. (CO1)

3. (a) Explain any *one* solid state transducer in detail. (CO2)

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(2)

OR

- (b) A common base phototransistor has input impedance of 20 ohm and output resistance of 100 kilo ohm. If a signal of 250 mV is applied between emitter and base, find the voltage amplification. Assume $\alpha = 1$. (CO2)
4. (a) Explain Piezoelectric effect in detail. What is the difference between piezoelectric and pyroelectric effect ? (CO2)

OR

- (b) A piezo crystal has the dimension 7 mm $\times$ 6mm $\times$ 1.5 mm. The force acting on it is 7 N. The charge sensitivity is 140 pC/N and its permittivity is 12.5×10^{-19} F/m. Calculate voltage generated and charge. (CO2)
5. (a) What is the working principle of Electromechanical transducers ? (CO2)

OR

- (b) Describe working of cantilever-based transducers. How will maximum deflection be calculated ? (CO2)